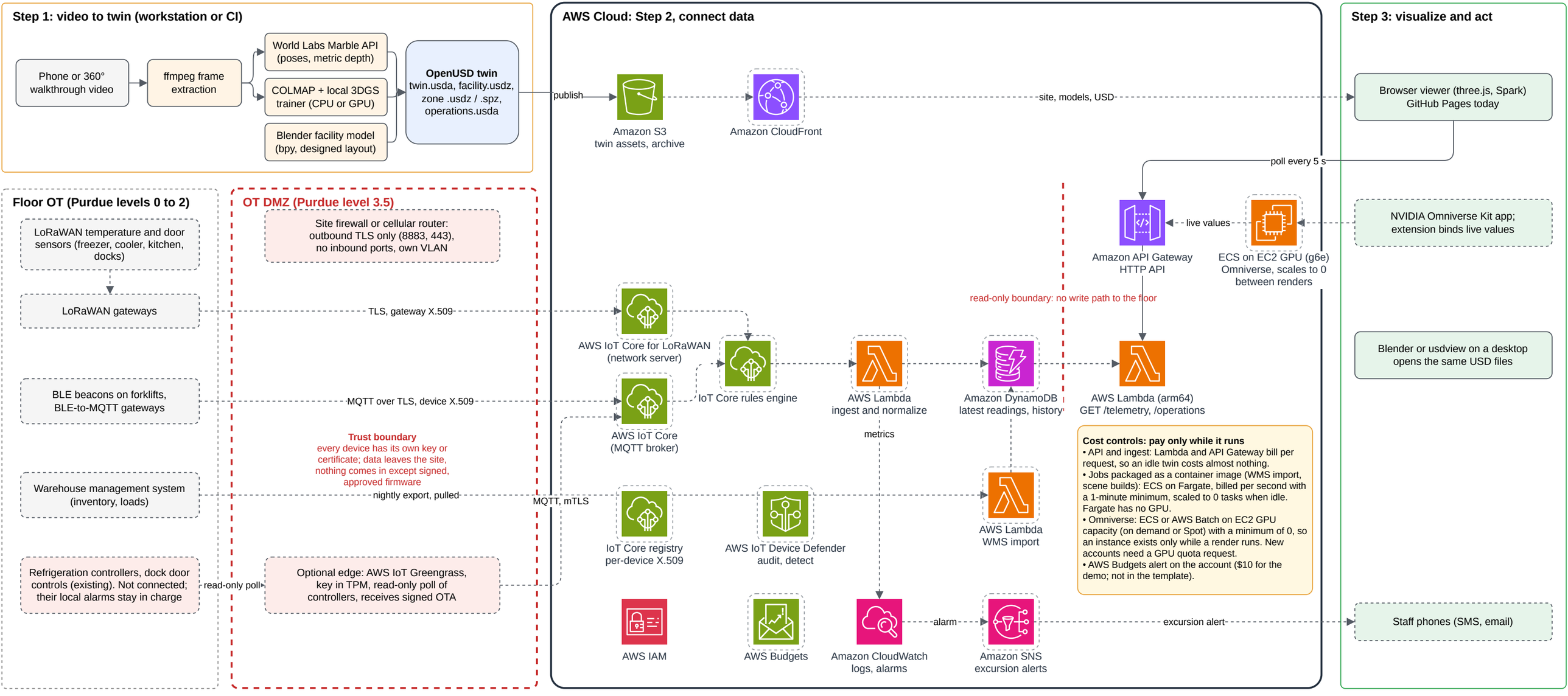
Food bank digital twin on AWS Cloud



OT security: zones, conduits and what is built (docs/OT_SECURITY.md)
Field devices and edge gateways sit on the floor; the OT DMZ passes only outbound TLS; cloud ingest accepts data and sends no commands; the twin is read-only. Existing refrigeration and dock controls are not connected. Red dashed lines are trust boundaries. The only designed path toward the floor is a signed, human-approved firmware update.
Built (AWS SAM template, simulated data): GET-only HTTPS API, throttled to 5 requests/s (burst 10), function role that can only write its own logs, no IoT credentials or device policies in the cloud, 7-day log retention.
Design only (dashed): per-device certificates and keys, least-privilege device policies, segmentation, the edge gateway, device monitoring, plausibility and heartbeat checks on readings, signed updates.

Solid: in the repository today and deployed in us-east-2 (AWS SAM template; simulated data). Dashed: real-floor path priced in docs/COST_ESTIMATE.md from public list prices; directional, enterprise pricing may differ. Red dashed: OT trust boundaries (docs/OT_SECURITY.md).